



Question 1

(a) Describe the advantages of storing lipids rather than starch in a seed. [2]

1. For the **same weight/mass**, lipid has a **higher** / twice the **amount of energy** than starch thus allowing lesser amount of fats to be stored for the same amount of energy;;
2. lipid hence **takes up less space/lighter** than carbohydrate which minimizes the mass of seed, especially important for seeds disperse by wind;;

OR

3. Lipid is **insoluble in water** and can be stored in large amount **without affecting water potential of cells**;;
4. So allows seed to be **stored for long periods of time** when conditions are dry and unsuitable for germination;; (1 mark each)

Comments: Some students fail to appreciate that answering “lipids store more energy than starch” is insufficient. It is important to note that lipids store more energy than starch **per unit mass**. Students are also advised to read the **context** of the question closely and exclude insignificant points which, while may be important in a different context or in an essay, are irrelevant to this question (storage of energy in seeds).

(b) Suggest with reasons where the hydrolytic enzyme is located in the aleurone cell. [2]

1. **Lysosome** since the hydrolytic enzyme causes breakdown of lipids;;
2. It is stored in a **membrane bound organelle** in the seed;
3. Kept **separated** from the food source/ lipids;
4. Until it is **required during germination**; (pt 1: 1 mark; pt 2 – 4: ½ each)

(c) Explain what determines the precise position of these three amino acids in the structure of the hydrolytic enzyme. [3]

1. enzyme has a **tertiary structure**;
2. **3-dimensional folding** (of the enzyme brings these amino acids into close proximity);
3. This is determined by the **primary structure** which is the specific **number, sequence and type** of amino acid along the polypeptide chain;
4. It results in formation of **hydrogen, ionic and disulphide bonds and hydrophobic interactions**;
5. between **R groups of amino acid far away**;
6. bringing these **distantly separated** amino acids **together**; (½ mark each)

Comments: Many students misinterpreted this question and discussed transcription and translation instead. A closer look at the question reveals that a better approach is to link the **primary** structure to the **tertiary** structure. “determines the precise position... in the structure of an enzyme” calls for an explanation of how amino acids far away in the amino acid sequence are brought close together in a specific location.

(d) Account for the curve at 15 °C in **Fig. 1.1**.

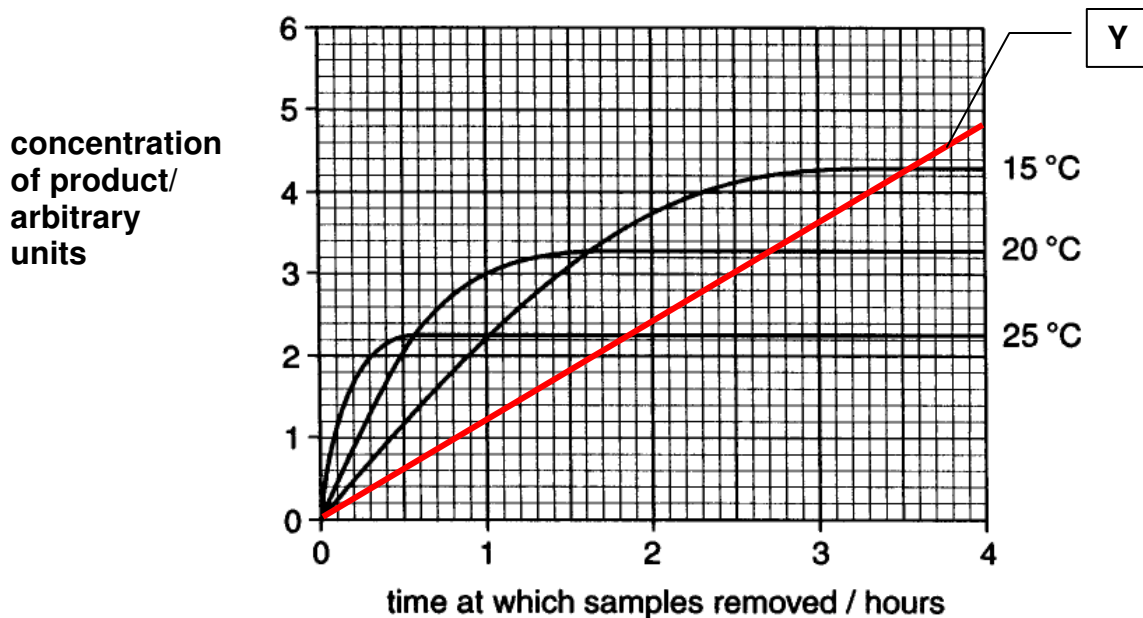
1. **steep increase** for the **first 1 – 2 hours / till 2.2 - 3.8 a.u; A linear, steady**
2. as enzyme is working / rate of reaction is high / reaction is proceeding;
3. substrate is converted into product;
4. curve has **levelled off** / became **constant** at **3 hours/ 4.3 a.u**;
5. after the enzyme was **denatured**;

6. because at 15 °C, kinetic energy supplied to the enzyme causes the **ionic and hydrogen bonds** to be broken; (½ mark each)

Comments: Generally poorly answered even though it is a simple question. A comparison between the various graphs is uncalled for. If the question requires a comparison, it will read something like “compare and account for...” or “with reference to the other graphs, explain...”. Let’s look at the command word “account”: it requires you to do two things i) describe the shape of the graph (ans pt 1 and 4), and ii) provide an explanation for this shape (ans pt 2, 3, 5, 6).

Feedback inhibition or active site saturation is not accepted as possible explanations for why the curve of 15°C plateaus at the third hour. Instead, denaturation of the enzymes is the only acceptable answer. The question “seed germinates best... at close to 5°C” indicates that 5°C is close to/is the optimum temperature. As such, given that the substrate concentration is not a limiting factor, the curves should plateau off at the same product concentration **if denaturation did not occur** at 15°C, 20°C and 25°C. This is evidently not so in Fig. 1.1 where we can see that the higher the temperature, the less the product formed. However, **if denaturation occurred**, we should see a reduction in product formed at higher temperatures, which is in line with the results shown in Fig. 1.1.

- (e) Predict and explain the effect of carrying out the procedure at 5 °C. Sketch your prediction on **Fig. 1.1**. Label it as Y. [1]



- rate of reaction is slower because **lower kinetic energy** compared to the other temperatures;
- less **effective collisions** between enzyme and substrate;
- resulting in less **enzyme-substrate complexes** formed and hence product formed per unit time;
- concentration of product is higher than for the other temperatures because it is the **optimum temperature** at which the hydrolytic enzyme works;
- as enzyme is not **denatured** at 5 °C;
- Q10 = 2**; (1/2 each, max. 2)

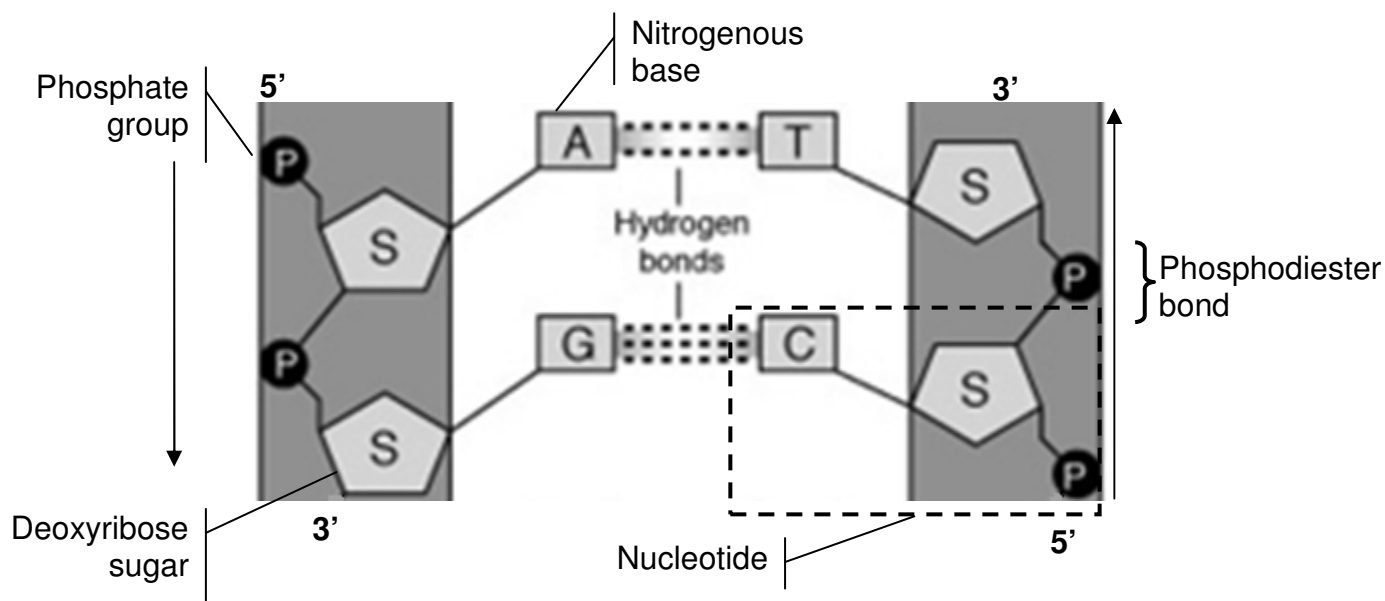
Comments:

Graph Y: i) Do not forget your SPA skills, use a **ruler** and **fine pencil** in drawing a graph. Remember to **label** your graphs too. Marks are not penalized for this exam but your mileage may vary for the A levels.

ii) Some students drew an S-shaped graph; unfortunately, this is not accepted as it does not follow the trend of the other graphs at 15°C, 20°C and 25°C. iii) Observant students, who will score the full one mark, will make an effort to show their understanding of Q10=2, for example, by ensuring that graph Y will cut the point where **[product] = 1.1a.u.** when **t = 1h** because the graph of 15°C cuts the point where **[product] = 2.2a.u.** when **t = 1h**)

Question 2

(a) (i) In the space below, draw a labeled diagram to show a representative structure of an exon. [4]



1. Phosphate group;
2. Deoxyribose sugar;
3. Nitrogenous base;
4. Correct pairing of A to T and G to C;
5. Correct no. of hydrogen bond between bases;
6. Correct labeling of 5' to 3' direction;
7. 2 strands of DNA;
8. Antiparallel arrangement;
9. Nucleotide;
10. Phosphodiester bond;

Students must draw a minimum of 2 base pairs;

Comments:

Some students do not know how that the representative structure of an exon is a DNA structure.

(b) Explain what happens if a mutation occurs at the splice site of the intron between exon 2 and 3. [3]

1. Small nuclear ribonucleoproteins (snRNPs) will not be able to recognize and bind to splice site of the intron on the mRNA;
2. **spliceosome** (pre-mRNA-snRNP complex) is not formed;
3. as the snRNPs cannot complex with one another to bring the upstream and downstream exons close together for splicing;
4. Thus the intron is not excised out;

5. and the mature mRNA formed is longer than normal;
6. to be translated to a much longer polypeptide than usual;
7. which will not be folded to the same 3D conformation as the normal protein;

Comments:

Reference to spliceosome not able to recognize splice site is rejected because the spliceosome is not formed in the first place. Other students have the misconception that the since intron is non-coding then the protein synthesized will not be affected.

(c) Eukaryotes can regulate their gene expression at various levels whereas prokaryotes regulate gene expression predominantly at the transcriptional level.

Explain why prokaryotes regulate gene expression predominantly at the transcriptional level. [2]

1. due to absence of a membrane-bound nucleus;
2. which means that transcription and translation occurs simultaneously in prokaryotes;
3. hence by regulating transcription, the bacteria is also regulating translation;
4. a need to regulate gene expression from the start/ at the earlier process;
5. genes are arranged in the form of operons in prokaryotes;
6. transcriptional control allows many genes to be regulated at the same time;

Comments:

Predominantly means mainly but does not imply that there is no translational control. It is however inappropriate to mention post-transcriptional modification here since prokaryotes do not have RNA processing. Reference to absence of histones so no regulation at chromosome level is also rejected as one can see that it is still transcriptional control for eukaryotes and hence not answering the question.

Question 3

(a) Explain the importance of this stage in the production of identical daughter cells. [2]

1. Stage = **DNA replication**;
2. DNA that is **identical**;
3. **Binary fission/ Two** daughter cell are produced
4. will each **inherit** the identical DNA;

(b) Using your understanding of the *trp* operon,

(i) describe the organization of the *his* operon. [3]

1. **Structural genes** that synthesize the enzymes for the synthesis of histidine are **clustered together**/ found next to each other;;
2. Under the **control** of a single **promoter**; and a **operator**;
3. Promoter and operator on the upstream of the structural genes; promoter followed by operator, then the structural genes; (idea of where promoter and operator are in relation to the structural genes)

Comments: Some students struggled to answer this question and many went out of point. This question asks about the **organization** of the operon and this requires you to discuss i) how the various components of the operon are related (ans pt 1 & 2), ii) and the **relative position** among the various parts of the operon (ans pt 3). Many students also made conceptual errors e.g. forgetting that both the promoter and operator are needed to control the expression of the structural genes, or giving the wrong position of the various components of the operon.

(ii) discuss how the *his* operon can be regulated by histidine concentration. [4]

1. Low [histidine], repressor is inactive;
2. and cannot bind to the operator;

3. RNA polymerase binds to the promoter;
4. and transcribes the genes for histidine biosynthesis;
5. High [histidine] – histidine act as a co-repressor;
6. and binds to the repressor;
7. This changes the conformation of the repressor to make it active, it binds to the operator;
8. Blocks the binding of the RNA polymerase/ prevents the RNA polymerase from transcribing the structural genes.

(c) Discuss the significance of having both inducible and repressible operons in the bacteria. [3]

1. Allows bacteria to **regulate gene expression** according to the **environment**;
2. Inducible operons – **enzymes synthesized only if substrates are available**;
3. Repressible operons – **enzyme production repressed if the product is available**;
4. **Saves energy and material** in not synthesizing extra enzymes that are not needed;;

(d) In eukaryotes, the insertion of viral DNA can cause tumor formation. Suggest a reason why such abnormality is not seen in the insertion of lambda phage into bacterial cells. [1]

- Bacteria are **single celled organism/unicellular**, any dysregulation will result in abnormal cells that will still exist as 2 individual cells, so no tumor formation.
- No cell cycle for bacteria as there is no mitosis;

Comments: Only a small minority of students got the right answer. The focus here is that bacteria are **single cell organisms** while tumors exist as a **mass of cells**. Thus, even if an insertion of viral DNA causes the bacteria cells to divide abnormally, they will not come together to form a tumor (or a mass of cell), but remain as separate rapidly dividing cells.

General comments for Q3: Many students can benefit from using the proper terms. Phrases like “control the gene” or “change protein production” could be improved by using terms like “regulate/control gene expression”, “upregulate” and “downregulate”.

Question 4

(a) (i) State and explain one source of ‘external stress’ that may trigger the activation of p53. [2]

- Exposure to UV light / radiation or oxidative stress or chemical mutagens / carcinogens;;
- Causes damages / changes to DNA;
- Transcription factor p53 activated to regulate transcription of genes that carry out repair of damaged DNA or cause apoptosis for damaged cell;

Comments:

Reject: errors in replication, failure for chromosomes to separate properly. Question is on external stresses. Answers should highlight causes extrinsic to the cell / tissue.

(ii) With reference to p53, explain the interaction between a transcription factor and control element. [3]

- p53 acting as the transcription factor (accept either activator / repressor);
- Binding specifically via the DNA binding site on the TF;
- To enhancer / silencer (corresponding answer to 1st point);
- Looping mechanism / bending of the DNA;
- Brings p53 to close proximity to transcription initiation complex;
- Interaction with RNA polymerase / transcription initiation complex;
- (Alternative) Prevent binding of RNAP to complex for repressor only;

- Up-regulation / down-regulation of transcription of target genes (corresponding answer to 1st point)

Comments:

Reject: p53 binding to promoter / p53 acting as basal transcription factors. It was given in the question that activated p53 binds to sequence located **far away** from the gene.

Reject: general statements describing what control elements and transcription factors and giving examples **without referring to p53 as the example of transcription factor** as stated in the question.

Reject: Details of various mechanisms in which silencer down-regulates / represses transcription. This is not required, as the emphasis should be the interaction between the TF and control element.

Reject: Explanations of the functions of p53 proteins eg. Transcription of target genes causes apoptosis / DNA repair etc. Not relevant to the question.

Answers should be focusing on the **interactions** between the transcription factor and control element in order to bring about the control on transcription.

(b) (i) With reference to Fig. 2, describe the effect of N-phosphorylation on the activity of Tac1 protein compared to control during the first 6 minutes. [3]

- Greater / Continual increase for N-phosphorylated protein compared to control;
- QV – from 0 to 62au for N-phosphorylated;
- QV – from 0 to 24au for control; and stayed relatively constant throughout;
- Lowered increase for C-phosphorylated protein compared to control;
- QV – from 0 to 13au for C-phosphorylated; and remained relatively constant throughout; (max 3)

Comments:

Reject: Answers that merely describe the 3 graphs separately without any comparisons. Note that the comparison should be done for N-phosphorylated *and* C-phosphorylated vs the control.

Reject comparison of N-phosphorylated vs C-phosphorylated proteins with no comparison made against control

In general, rather straightforward data analysis question, except for a small group of students who mistakenly took the C-phosphorylated graph as the control graph. Generally most students are quoting the correct values. Reminder to include units as well for some students.

(ii) Suggest what could have happened to lead to a decrease in N-phosphorylated Tac1 activity after the 6th minute. [2]

- Negative feedback;
- Removal of stimulus / external stress;
- Addition of ubiquitin to protein;
- Mark it for degradation by proteasome;
- Enable cell to dispose of proteins that have completed their activity;
- (Alternative) phosphatase removing phosphate from protein;
- protein revert to activity level of control;

Comments:

Reject: inorganic phosphates are used up

Note the appropriate use of terms: Proteins are degraded, not hydrolysed. Bonds are hydrolysed.

(iii) It was discovered that the phosphorylation to the C-terminus limits the ability of the Tac1 protein to bind to DNA. Suggest how this may occur. [2]

- Adding phosphate group may lead to changes in R-group interactions within the Tac1 protein;
- Changes in tertiary structure / 3D conformation of protein;
- Protein binding site no longer complementary to DNA;
- DNA-binding site of protein at the C-terminus of protein;
- Phosphate group may block the binding site of protein OR create steric hindrance;
- (Alternative) Phosphate group being negatively charged would repel the negatively charged DNA, hence lower affinity of protein to DNA;;

Reject: idea of non-competitive inhibition by the phosphate group. Note that Tac1 is not an enzyme.

Comments:

There is a tendency of students to confuse proteins with DNA in this question.

Some example: There was the mentioning of free 3'OH availability of the protein for it to bind to DNA.

This is incorrect. In addition, DNA does not bind to the '*active site*' of the transcription factor. It should be the DNA-binding site instead. There were also incorrect usage of the term 'complementary base pairing' to describe the binding of the protein to the DNA. This is only applicable when both sequences are nucleotides.

Question 5

(a) Explain what is meant by multiple alleles? [2]

1. The gene has more than 2; (@ multiple) alternative forms (of a gene) (@ alleles);
2. causes slightly different phenotypes for the same characteristic;
3. occupy the same gene locus on homologous chromosomes;
4. At any one time, a diploid organism individual can only have 2 alleles;

NB: Points 1 and 2 must be present to get the full mark.

Comments: When asked to explain certain terms, students should not reused the words that are given eg. saying that "multiple alleles" means many *alleles* or *multiple* alternative forms of a gene. Both terms have to be explained.

(b) Use one piece of evidence shown in Figure 5 explain clearly why the blood type for **individual 4** (shaded circle) might have been wrongly typed. [2]

Either

1. Mother is blood group AB – can only be $I^A I^B$; (1/2m)
2. Father is blood group A – can be $I^A I^O$; (1/2m) becos 4's brother is blood group B;
3. Their children can only have blood groups A ($I^A I^O$ or $I^A I^A$), AB ($I^A I^B$) or B ($I^B I^O$) – correct blood groups with genotypes;;
(½ m each for the correct blood groups and their genotypes)

Or

1. If individual 4 has blood group O, she can only be $I^O I^O$;
2. Her husband can be $I^A I^O$ or $I^A I^A$;
3. Her children must then inherit an I^O allele from her;
4. Her children can only have blood groups A or O/ Her children cannot have blood groups AB ($I^A I^B$) or B ($I^B I^O$) since the allele I^O is recessive to both I^A and I^B ;

Comments: It is disappointing that quite a number of students did not use the standard form of representing the alleles for the ABO blood group. These are I^A , I^B and I^O . Since this question asked for explanation, re-stating what is obvious in the pedigree will not gain any mark eg. Individual 4 cannot have blood group O because her father and mother has blood group A and AB respectively. Genotypes of these individuals have to be given in the explanation. Many answers that just state the genotypes of the individuals also were not given full mark. An explanation that blood group O requires an $I^O I^O$ genotype as the allele I^O is recessive to both I^A and I^B is necessary to make the answer complete.

(c)(i) State the importance of the antigens on the RBC. [1]

For cell recognition i.e for the immune system to be able to differentiate between cells that belong to itself and cells that are foreign/ self and non-self;;

Comments: Antigens present on cell surface serve the function of cell-to-cell recognition and cell-to-cell adhesion. In this context, the first function is the answer. Antigens present allow the immune cells in the body to recognize self from non-self. If the cells carry antigens that are recognized as foreign by the immune system, the immune cells will mount an immune response which basically means the foreign cells will be destroyed.

(ii) Suggest a possible mode of interaction between the two gene loci (i.e. H/h and $I^A/I^B/I^O$) and explain for individual 4's phenotype. [3]

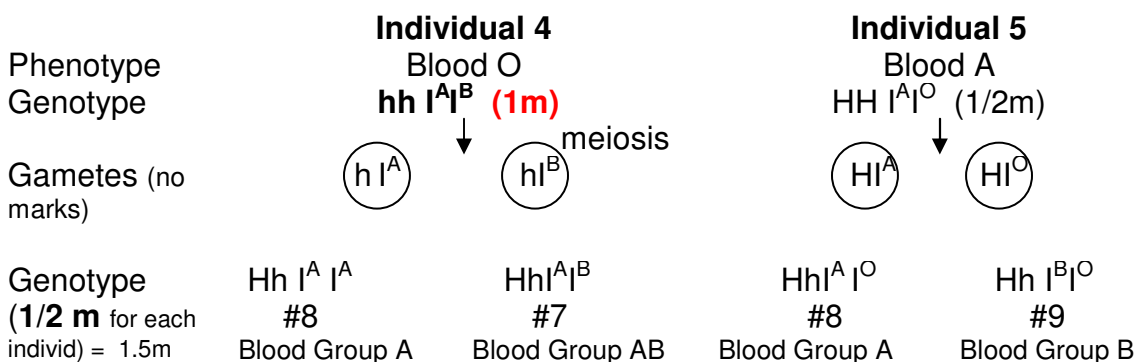
1. Recessive **epistasis**;
2. Allele H – allows for the synthesis of a compound that is important in antigen expression/ synthesizes an inhibitor that prevent antigen production;
3. Individual 4: hh/ homozygous recessive; (because she is the only person with a discrepancy in the blood type)
4. no antigen present on the RBC;
5. (so the lack of antigen –)she is typed as O/ phenotypically blood group O;
6. regardless of the type of alleles that she has in the locus for blood group/ hh mask the expression of the alleles at the $I^A / I^B / I^O$;
Accept the alternative: If those with HH/Hh, there will be antigen production, so allowing the expression of the corresponding antigen found on the ABO locus
7. Parents of Individual 4 must be heterozygotes Hh to produce a homozygote hh;

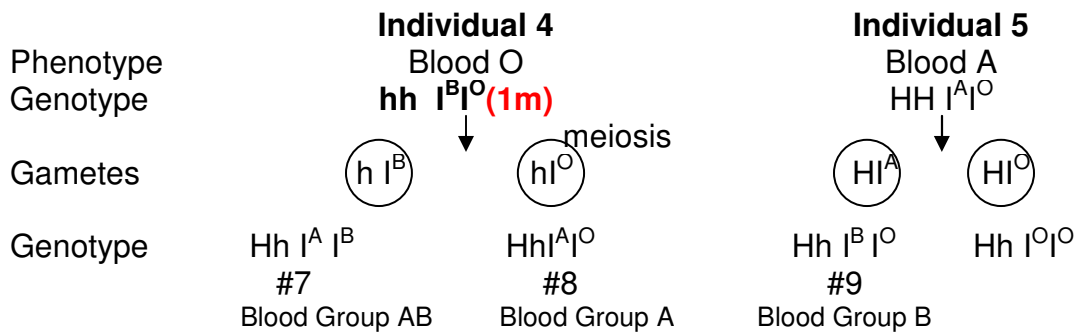
Comments:

The genes cannot be linked as there is only one characteristic that we are dealing with – that is blood group. In linked genes, there should still be two phenotypes that are expressed. In describing individual 4, students should note that the **genotype of 4 is not changed**. It is her **phenotype** that is 'changed'.

(iii) Draw a genetic diagram to explain one possible way in which individuals 7, 8 and 9 could inherit their blood types form their parents. [3]

Either one below:





Comments:

Some students did not relate the genotype to the phenotype (i.e. blood group) and the individual who is expressing the blood type (eg. individual 7 – relate to blood group AB).

Question 6

(a) (i) Explain the function of the myelin sheath. [max 2]

1. Hydrophobic in nature / consists mainly of lipids;
2. Prevents the direct entry of ions across the membrane;
3. Allows for saltatory conduction/ nerves impulses jump from node to node;
4. **Limits changes to membrane potential at the nodes of Ranvier;**
5. Increase the speed of transmission of impulses along the axon;

(ii) A person suffering from a rare, neurological disease has neurons that have degenerating myelin sheath. Suggest a possible symptom and explain how this may arise in this person.

1. Slow mental processing / delayed response/difficulty in coordinating movement;
2. (Loss of myelin sheath on the neurons) hence unable to conduct saltatory conduction/ more 'nodes' formed at regions where degeneration occurs;
3. Action potential/ depolarization can only be generated at adjacent/neighbouring areas;
4. Nerves transmission along axon is slowed down;

Comments:

Reject any answer that seems to indicate that the 'problem' is with the muscle eg. muscle do not contract as fast.

(iii) The degenerated neurones now have the axon membrane exposed to the surroundings. Explain how the **molecular structure** of the axon membrane facilitates nervous transmission along the axon. [4]

- Phospholipid bilayer;
- with a hydrophobic core;
- Prevents direct entry of charged ions Na^+ , K^+ into the axon;
- Allows accumulation of ions on one side of the membrane;
- Proteins distributed on the membrane;
 - Leak channels for Na^+ and K^+
 - Always open and allows ions to move down concentration gradient by *facilitated diffusion*;
 - Helps maintain resting potential (with Na^+/K^+ pump);

Marked under L; max 1m

Marked under C; max 1m

- Voltage-gated channels for Na⁺ and K⁺ ions;
 - Acts as channels for *facilitated diffusion* of ions down the respective concentration gradients of ions;
 - Opening of Na⁺ channels allows Na⁺ influx and depolarization of membrane resulting in an action potential;
 - Opening of K⁺ channels allows K⁺ efflux and repolarization of membrane;
- } Marked under V; max 1m
-
- Na⁺/K⁺ pump;
 - Actively transport ions against concentration gradient/ 3 Na⁺ out and 2 K⁺ in;
 - Maintains the resting membrane potential / maintain unequal concentration gradients of Na⁺ and K⁺ across the membrane;
 - Along with the selective permeability of membrane to K⁺ over Na⁺ ions; OR supported by more leak channels for K⁺ present than Na⁺ leak channels on the membrane;
- } Marked under P; max 1m

(b) Explain why transmembrane proteins are especially suited for a role in cell signalling. [3]

General to cell signaling:

- **Spans the entire membrane;**
- with one part interacting with **extracellular** ligand/ signal molecule;
- and another interacting with **intracellular** proteins;
- ligand binds via complementary fitting;
- Leading to conformational change of the transmembrane protein/ dimerisation of receptor;
- Cause activation of intracellular enzymes / intracellular proteins inside the cell;
- Allows cellular responses without the external signal having to move into the cell/ able to transduce external signal into internal signal;

Or specific to nervous transmission

- **Spans the entire membrane;**
- with one part interacting with **extracellular** ligand/ signal molecule;
- ligand binds via complementary fitting;
- Leading to conformational change of the transmembrane protein resulting in the opening of a hydrophilic channel (within the protein);
- Allows the diffusion of ions into the cell to bring about a cellular response;

Question 7

(a) What do you understand by the term “species”? [2]

- **Population** of organisms;
- That can **interbreed** with one another;
- To form **viable fertile** offspring;
- But do not interbreed with members of other species;
- Found in the same geographical location;

Comments:

Given number of marks and context of this question, students should focus their answers on the biological concept of species only.

(b) With reference to **Table 7.2**, describe and explain the evidence that *D. miranda* may be more closely related to *D. persimilis* than to *D. pseudoobscura*. [2]

- Less divergence for *D. persimilis* from *D. miranda* compared to *D. pseudoobscura* from *D. miranda* or more similar DNA sequence for *D. persimilis* and *D. miranda*;
- QV;
- Share a more recent common ancestor / diverge from a common ancestor more recently;
- So less time to accumulate mutations / less mutations observed in the DNA sequences;

Comments:

Some students chose the wrong values to quote in their answers. Note that the values chosen should be coming from the same region so that the comparison is fair.

(c) Of the 4 regions that were used in this study, a high proportion of them were non-coding DNA. Explain why non-coding DNA is suitable for this purpose.

- Non-coding DNA are not expressed in phenotype / do not code for protein product;
- Changes in non-coding DNA is neutral;
- Hence they do not affect the fitness of the organism;
- And are not subjected to natural selection;
- Changes in non-coding DNA can be accumulated;
- Can be used as molecular clock as accumulation may be more constant;

Comments:

Students have the following misconceptions –

There is a higher frequency of mutations occurring in non-coding DNA as they do not alter the fitness of organisms.

Coding DNA codes for genes that give rise to important protein products, hence mutations takes place slower.

Note that mutation is a random event that should have an equal chance of occurring at the DNA regions. ½ mark is deducted if species name is not underlined.

(d) Discuss the importance of this variation if it occurs in the coding region. [3]

1. Variation in coding region – codes for new product / phenotype or create new alleles in gene pool;
2. Increase raw material for natural selection;
3. Organisms with the most favourable combinations of alleles / characteristics selected for;
4. Higher chance of surviving, reproducing and passing on the allele to the offspring;
5. Allows better adaptation to the environment;
6. In the event of a sudden / drastic change;
7. Prevent the extinction of the entire species / better survival for species;

Comments:

Most students did not address the question fully. The question is on the **importance** of variation in coding region. Most students have no problems identifying the consequences of variation occurring (points 1 – 4) but usually fall short of stating the importance ie. adaptation to changing environment (points 5 – 7). Degeneracy of the genetic code leading to the same a.a coded for does not really address any significance from variation in the coding region, since the a.a has no change in identity.

- (e) If a new island were to emerge from the ocean as indicated in the diagram marked X, predict what the *Drosophila* population on the island will be like after 1 year. Explain your answer.

(You are to assume that *Drosophila* is the only insect that inhabits this island and that they have a generation time, i.e. from egg to adult, of about 10 days.) [2]

1. Similar to *D.persimilis* and *D. Miranda*;
 2. Ancestor migrated from the nearest / closest mainland;
 3. Divergence / adaptive radiation;
 4. As they explore new ecological niches or subjected to different selection pressures;
 5. Geographical isolation / no gene flow;
 6. Formation of other new species on island X;
- (Alternative) Founder's effect;

Reject: Population to be made up of *D.persimilis* and *D. miranda*.

Given the time span, they are likely to have diverged into other species that are similar to them.

½ mark is deducted if species name is not underlined.

Essay Questions

Question 8

- (a) Describe starting from mature mRNA, how a glycoprotein is synthesized and embedded in the cell surface membrane. [10]

R: During Translation, triplet base sequence of mRNA directs the ordered polymerisation of specific amino acids to construct the protein;;

Involves

Preparation (max 1) **P**

- Amino acid activation occurs;
- Attachment of a specific amino acid to the 3' end of the tRNA;
- Catalysed by specific amino-acyl tRNA synthetases whereby energy in the form of ATP is required;
- Aminoacyl-tRNA complex is formed;

Initiation (max 2) **I**

- The initiator tRNA, carrying the amino acid methionine (Met) binds to the small subunit of the ribosome;
- Finder sequence on 5' end of mRNA binds to antifinder sequence of rRNA at the mRNA binding site on the small subunit of the ribosome;
- Small subunit with the initiator tRNA bound moves downstream along mRNA until it reaches the start codon (AUG) on the mRNA and complementary base-pairing via its anticodon, UAC on the initiator tRNA occurs;
- the process is followed by the attachment of a large ribosomal subunit with hydrolysis of GTP,
- to form a translation initiation complex;
- At the completion of the initiation process, the initiator tRNA is positioned at the peptidyl-tRNA binding site (P site) of the large ribosomal subunit;
- and Amino acid-tRNA binding site (A site) of the large ribosomal subunit is ready for the next aminoacyl tRNA

Elongation (max 2) **E**

- Codon occupying the A site binds amino acid 2-tRNA with complementary anticodon;
- enzyme peptidyltransferase / peptide synthetase found in large subunit of ribosome catalyzes
- peptide bond formation between amino acid 1 (methionine) occupying P site and amino acid 2 occupying A site
- dissociation of amino acid 1 (methionine) in P site from its tRNA
- Meanwhile, there is **translocation** of ribosome **3 bases** down in the **5' to 3' direction of the mRNA**
- Dipeptide-tRNA complex now sits on the **P site** and a new codon is exposed at the free **A site**, free tRNA exits through **E site**
- An amino acid 3-tRNA molecule with complementary anticodon will come and occupy the free A site. The process is repeated to assemble amino acids in the correct order as the polypeptide chain elongates

Termination (max 1) **T**

- When a stop codon e.g. UAA, UAG, UGA occupies A site / reading of stop codon,
- release factor binds;;
- The release factor causes the **addition of a water** molecule instead of an amino acid to the polypeptide chain
- bond between final amino acid and tRNA is hydrolysed and
- dissociation of translation complex/ ribosome dissociates from mRNA

Rough endoplasmic reticulum rER (max 2) **R**

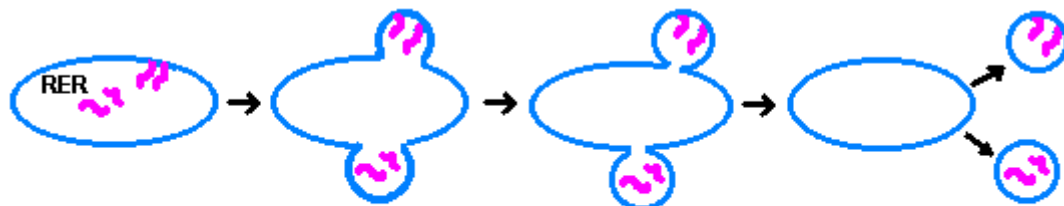
- polypeptide synthesis always begins in the cytosol, when a free ribosome starts to translate an mRNA molecule;
- the polypeptide destined to be embedded in the cell surface membrane is marked by a **signal peptide**, which targets the protein to the ER
- The signal peptide binds to a **signal recognition particle (SRP)**, which prevents further synthesis.
- The entire complex migrates to and attaches itself to a **receptor protein** in the ER membrane/ idea that the ribosome is attached to ER, i.e., translation takes place on rough ER;
- (The signal recognition particle dissociates off and protein synthesis continues)
- with the **polypeptide is released from the ribosome directly into the lumen of the ER**
- polypeptide chain **folds into its specific shape**;
- with the help of **chaperone proteins**;
- and **glycosylated**/ carbohydrates covalently bonded/ attached to protein;
- As it is a membrane protein, it **remains partially embedded in the ER membrane**;
- **transport** (R: secretory) **vesicle** leaves rough endoplasmic reticulum, carrying the glycoprotein in its membrane;

GA (max 2) **G**

- Transport vesicle fuses with cis face of GA which is located near ER;
- Glycoprotein undergoes **further chemical modification**
- during transit from cis to trans region of GA
- e.g. Golgi enzymes modify the **carbohydrate chain** of the **glycoprotein** (1/2)
- **sorting + packing** of glycoprotein into membrane of GA
- **and pinched off** from the **trans face** of Golgi apparatus as **Golgi vesicle**

Others O

- migrate **towards the cell surface membrane**
- Membrane of the vesicle **fuses** with the cell surface membrane
- Ref to **inward arrangement** of the **carbohydrate chain of glycoprotein** in the vesicle membrane
- results in **glycoprotein at the outer surface of the cell membrane** / glycoprotein points outward of cell after fusion occurs



Transport of membrane bound and free proteins from the lumen of RER to golgi in transitional vesicles

Comments:

1. In general, question is well done with some students scoring maximum in the different sections, especially in the translation process.
2. Some students have not read the question carefully and describe RNA processing and even transcription. This results in insufficient time for other sections of the paper.
3. Most students know translation well although there are some who are confused with translation in prokaryotes hence mention of 50S and 30S ribosomal subunits, formyl-methionine, and sequence of events in initiation of prokaryotes. As question is '... starting from **mature mRNA** ...' indicates that it is referring to a eukaryotic cell.
4. Some students failed to mention that translation takes place on rER and account indicates that polypeptide is synthesized in cytoplasm and directly transferred to GA.
5. Description for rER is sometimes vague for a number of students as this requires link made from various topics such as membrane and organelles, protein synthesis, genetics of viruses, etc. More marking points are given here to credit those students who have made the attempt to put in relevant points such as mention of **signal peptide**. Again a maximum of 2 marks is given here so that a balance description is achieved.
6. Note that vesicle pinched off from GA is not necessary secretory vesicle as in this case, the glycoprotein is not secreted out of the cell.

(b) Explain how HIV and influenza virus can cause diseases in animals. [5]

HIV virus (H) (max 3)

1. HIV targets lymphocyte cells such as the **T helper cells** by binding its glycoprotein **gp120** to specific receptor sites (e.g. **CD4 receptor**) on the cells and enter via fusion;;
2. Damage to T helper cells which are involved in **establishing and maximizing capabilities of immune system**/ activating and detecting other immune cells leads to a **weakened immune system**
3. HIV-infected patient will not be able to defend against **opportunistic infections** by other common viruses and bacteria leading to **acquired immunodeficiency syndrome (AIDS)** which includes pulmonary infections (e.g. pneumonia) and gastrointestinal infections (e.g. chronic diarrhea) in the patient.
4. The **insertion of the retroviral DNA into the host genome** may also cause **cancer** due to **disruption of tumour suppressive genes** or **conversion of proto-oncogene to oncogene**.

5. HIV may also induce **adjacent host cells to fuse together** forming **giant multinucleated cells** or syncytia allowing the HIV virus to **kill many lymphocytes** by infecting only one.
6. Moreover, HIV has a high rate of mutation and its intracellular recombination with other strains of HIV produces altered gp120 to which antibodies made against the earlier strains of HIV can no longer bind.

Influenza virus (I) (max 3)

1. Influenza virus targets **epithelial cells of respiratory tract** by binding its **hemagglutinin** onto **specific receptor sites** (or sialic acid sugars) and enter via endocytosis;;
2. Loss/Damage of epithelial cells results in **impairment of the respiratory tract function** e.g. impair **upward movement of mucus** by **ciliated epithelial cells** that line the respiratory tract;;
3. **Reduced clearance of infectious agents** from respiratory tract leads to symptoms like chills, fever, sore throat, muscle pains, headache, coughing, pneumonia/ difficulty in breathing/ inflammatory condition of the lung (any 2 symptoms)

General G (max 2)

1. Within the host cells, the replication and assembly of the viruses involve **inhibit normal host cell DNA, RNA, or protein synthesis**;
 → **depletes the host cell of cellular materials** (e.g. amino acids, nucleotides) essential for life or normal function;
 → may also **cause structural or functional defects** in the infected host cell leading to **cytolysis or altered cell functions**;
 → any one
2. **Attachment of the viral proteins and glycoproteins on the surface of the infected host cell surface membrane** may result in:
 → infected host cell being **recognized as foreign** and **destroyed by the body's immune defenses**;
 → **infected cells releasing inflammatory cytokines** that will **induce an inflammatory reaction which damages other healthy cells**;
 → any one
3. The release of the matured viruses from the host cells by **budding** may lead to **death of the infected cells** due to **extensive loss of the plasma membrane**;;

Comments:

1. A straightforward question which can easily score full marks for students who have studied.
2. No major misconceptions here except for the type of cells and specific receptor sites, and the glycoproteins of viruses where some mix-up occur.

(c) Discuss the significance of the population (of a species) in evolution. [5]

1. Population is a group of **interbreeding** individuals;
2. sharing the **same gene pool**;
3. through interbreeding, **provides the genetic variation** for **natural selection to work on**;
4. via **meiosis** and **random fusion of gametes** during fertilization;
5. A population lives in the **same location/locality**;
6. and are subjected to the **same selective pressures/** environment conditions;
7. Those selected for will **survive, reproduce** and **pass on their favourable traits/** favorable combinations of alleles **to their offspring/ next generation**;
8. **In time**, there will be **changes in the allele frequencies**;

9. As it is the **collective genetic response** of the population that **determines survival of the species** and the **formation of new species**
10. Population is the smallest unit that can evolve;
11. As evolution can only **be measured as changes in relative proportions of heritable variations** in a population in **successive generations/ evolutionary impact of natural selection is only apparent in the changes of a population over time**

Comments:

1. Marking point is based on 1 mark so most students were able to do well in this section although there is a significant number who did not attempt this section. If the marking point is more stringent, then many would not have done as well.
2. Most students started off with population being the smallest unit that can evolve without defining the term population.
3. Point 2 is rarely seen and often used in the context of mutation serving to increase the gene pool which is not relevant here. For natural selection to work, there must be variation in the population.
4. Point 6: Some students have the misconception that it is **different** selection pressures operating but that is only applicable to different environment/ locality when the species are separated from each other.
5. Point 7: there is still a significant number who has 2 out of the 3 key terms here and half mark was given here, through leniency of the marker.

Question 9

- (a)** Explain the role of isolating mechanisms (eg. geographical, physiological and behavioural) in the evolution of new species. [10]

Comments:

Students could have started off the essay by framing their answers. Two important points to clarify. These are (i) *What are isolating mechanisms?* (ii) *How are new species form?*

*An isolating mechanism or a **reproductive barrier** is any factor that prevents two species from producing fertile hybrids, thus contributing to reproductive isolation. You will also realize that the species concept that is being used here is the biological species concept since reproductive barrier and fertile/viable offspring are being used.*

G Geographical isolation [5m max]

1. Allopatric speciation/ there is physical separation of 2 subpopulations/populations;
2. Example of a physical barrier: mountains/ oceans etc
3. Members of the 2 populations prevented from mating as they do not meet;
4. No gene flow between them;
5. Each environment has different selection pressures;
6. Mutations can also occur independently in each population;
7. Each population undergoes natural selection;
8. Overtime, genetic differences between the 2 populations increases/ both populations accumulates genetic differences;
9. Genetic drift can also occur in these populations;
10. The 2 gene pools diverge (reference to point 8);
11. (*Over time*) Members of the one population cannot interbreed with members of another population/ they are reproductively isolated from each other;

P Physiological isolation [4m max]

1m for an explanation of any of the follow mechanisms, capped at a max of 3m if students did not explain how these can bring about evolution of new species.

Gamete incompatibility/ Hybrid infertility/ Habitat differentiation/ Polyploidy/ Temporal isolation/ / Mechanical isolation;;

Example of an explanation:

Allopolyploid

- When 2 different species interbreed and produce a hybrid offspring, hybrid are often sterile;
- In some cases, these hybrid may undergo non-disjunction to form fertile/viable polyploids (=allopolyploid);
- These polyploids are then able to mate with each other but not with the parental species, thus they represent a new species;

B Behavioural isolation

1. mating rituals are different/ coloration differs/ different calls etc;
2. No attraction between males and females of different species;
3. Prevents interbreeding/ keep gene pool distinct/ prevents gene flow;

Others

1. Behavioural and Physiological isolation will bring about sympatric speciation;

Comments:

Many of the mechanisms under physiological and behavioural serve to keep species distinct. Therefore students must refer to how these mechanisms allow evolution to take place.

This will involve different selection pressures selecting for certain variants with combinations of favourable traits that are inheritable. With the selection of these characteristics, there likely will have to be some form of reproductive barrier so that eventually these group of variants are isolated from the main population.

(b) Compare and contrast the structure and function of collagen and cellulose. [10]

	Differences	Collagen	Cellulose
1	Located	Animals	Plants
2	Subunit	Amino acids	B-glucose
3	Type of subunit	Different types of a.a	One type of glucose monomer (B-glucose)
4	Bonds	Peptide bonds	Glycosidic bonds

5	Aggregation	Tropocollagen – fibrils - fibers	Micelles – microfibrils - macrofibrils
6	Nature	Polypeptide (Protein)	Polysaccharide (Carbohydrate)
7	Cross linkages	Covalent bonds between tropocollagen	Hydrogen bonds between cellulose chains
8	Natural uses	Connective tissues – ligaments, tendons, in teeth and bone formation	Material for cell wall
9	Elements	C, H, O, N, S	C, H, O
10	Insoluble because...	Large molecules that consists of many hydrophobic amino acids	No available –OH group as they are used for cross-linking H bonds
Similarities			
1	Solubility	Both are insoluble in water;	
2	Function	Both plays role in structural function; provide high tensile strength;	
3	Structure	Both have fibrous and rigid structure; No branching, linear chains;	
4	Cross-linkages and bonds	Cross-linking between chains; Hydrogen bonds between cellulose chains and between polypeptides in a tropocollagen;	
5		Both are macromolecules; Made up of many smaller subunits;	

(max 10marks)

Comments:

Collagen is not found *in* animal cells but rather in the extracellular matrix and connective tissues.

Note the appropriate scale of comparison. For instance, if you are comparing the bonds between the subunits, and listed *glycosidic bonds* for cellulose, then correspondingly it should be *peptide bonds* for collagen, and not *covalent bonds*. Another example - the monomer of collagen is not tropocollagen if you compare it to B-glucose in cellulose. It should be amino acid instead.

There are still students who incorrectly deemed cellulose to have secondary structure compared to quaternary structure for collagen. The structural denotation refers to proteins only.

No marks awarded for a similarity if one component is wrong.